SWE 307 BIG DATA PROJECT - 4 DESCRIPTION

RAG-AI Platform with vector DB

Due date: 10.12.2026 Thursday, in class.

In this project you are going to implement a platform that uses Retrieval Augmented Generation (RAG) technique that allows LLM models to pull current and specific information from external sources. In this study, you will use Ollama server and open source Llama 3.2 model (as you already used in Project 2) for information extraction. You are expected to create a RAG pipeline that includes extraction, chunking, embedding and ingestion to vector database (see Figure 1). Examples will be done in the class.

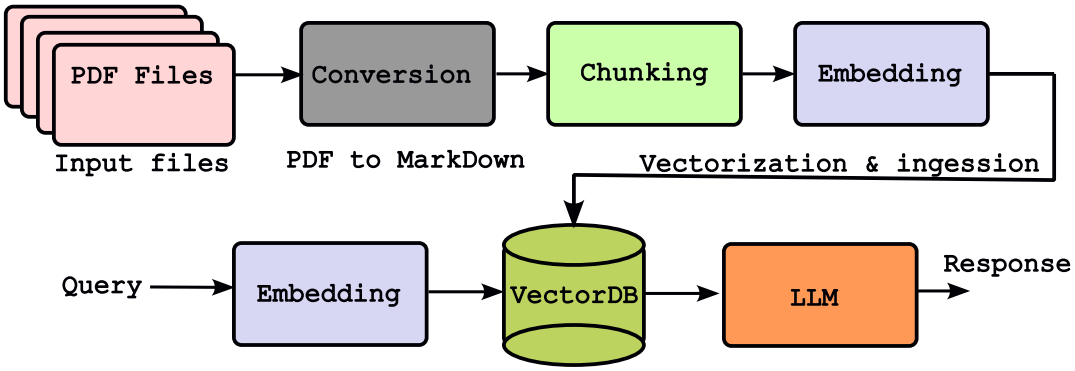


Figure 1. The architectural block diagram of the project 4.

Solution specifications:
1) PDF files: Use “SAU-EğitimÖğretimYönetmeliği.pdf” download it from https://kms.kaysis.gov.tr/Home/Goster/151552?AspxAutoDetectCookieSupport=1
2) Conversation: Use “Docling” from IBM.
3) Chunking: You can use “LangChain” or “LlamaIndex”.
4) Embedding: Use “qwen3-embeddings”.
5) VectorDB: Use “ChromaDB”.
6) LLM server: Use “Ollama”.
7) LLM model: Use “Llama3.2”.